Lab 8: Assumption-lean regression

Causal Machine Learning

Stijn Vansteelandt, Ghent University

The First Steps program was set up in 1989 in Washington State, United States, in order to serve low-income pregnant women and children via maternity care necessary to ensure healthy birth outcomes. A specific goal was to reduce the risk of low birth weight. Using data obtained from birth certificates from 2 500 children (singletons) born in King County, Washington in 2001, we will evaluate the 'effects' of the First Steps program on the risk of low birth weight (defined as birth weight below 2500 grams), as well as its association with maternal age. Read the data in R. Available data are (in order of appearance of the columns) the child's gender, whether it concerned a singleton birth or not, maternal age, race, parity, marital status, birth weight, participation in the First Steps program, participation in a welfare program, whether the mother smoked during pregnancy, whether the mother consumed alcohol during pregnancy, maternal weight before pregnancy (lbs), maternal weight gain during pregnancy (lbs), education (years), gestation (weeks).

1. Use assumption-lean linear regression to quantify the effect of participation in the First Steps program on the risk of low birth weight, adjusting for possible confounding. Use `SuperLearner` to estimate the unknown nuisance parameters. It may be wise to first do this without cross-fitting and for a small library of learners. Calculate a corresponding 95% confidence interval.

   (a) What do you think might be relevant confounders? If this is hard to judge, what question would you ask a subject-matter expert, knowing that the expert may not be sufficiently familiar with the concept of confounding?

   (b) Interpret the results, first pretending that the effect of participation in the First Steps program on the risk of low birth weight is constant. Does it equal an population-average treatment effect in that case?

   (c) It is unlikely that the effect of participation in the First Steps program on the risk of low birth weight is constant (why?). How does that affect the interpretation? (Note that the interpretation given in the previous question, even if not strictly correct, should be reasonably valid).

   (d) In what sense are your obtained results more reliable than those you could have obtained using standard linear regression (possibly with model building)?

   (e) If time allows, contrast your result with estimates of the average treatment effect obtained using AIPW or TMLE.

2. Statistical and machine learning analyses typically weigh different individuals unequally depending on how much information they provide for the analysis, but we usually don't know how the weighting is done. In contrast, assumption-lean regression methods are open about this: they 'weigh' study participants unequally by

   $$P(A = 1|L)P(A = 0|L)$$

   for exposure A and covariates L. For each baseline covariate $X \subseteq L$, it is therefore of interest to report as summary statistics the corresponding weighted average of X:

   $$\theta = \frac{E\{P(A = 1|L)P(A = 0|L)X\}}{E\{P(A = 1|L)P(A = 0|L)\}}$$

To compute this, it is not valid to simply plug machine learning predictions for the propensity score into the above expression!

(a) Briefly explain why.

(b) We must instead estimate θ based on its efficient influence curve, which is given by

$$-\frac{[P(A=1|L)P(A=0|L)+\{1-2P(A=1|L)\}\{A-P(A=1|L)\}](X-\theta)}{E\{P(A=1|L)P(A=0|L)\}}$$

Use this result to summarize the baseline characteristics in this weighted population, along with a standard error or 95% confidence interval.

3. Use assumption-lean linear regression to quantify the linear association between maternal age and the risk of low birth weight. Calculate a corresponding 95% confidence interval. Interpret the results in terms of a 'shift intervention'.

4. Improve your previous results by using cross-fitting and by extending the considered library in the `SuperLearner`.

5. Repeat the 2 analyses in questions 1 and 3, but now presenting results on the risk ratio scale.

6. Repeat the 2 analyses in questions 1 and 3, but now presenting results on the odds ratio scale. Note that odds ratios are more difficult to interpret than risk differences and risk ratios. An advantage of assumption-lean regression is that you can choose the scale that you like, since the statistical model is merely used as an approximation, not an assumption.

   (a) Even so, there can be advantages to reporting odds ratios? Which?

   (b) How would you make the results interpretable now, considering that odds ratios are difficult to interpret?